

2920/302
MANAGEMENT INFORMATION SYSTEMS
November 2017
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE III
MANAGEMENT INFORMATION SYSTEMS
3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of **EIGHT** questions.
Answer any **FIVE** of the **EIGHT** questions in the answer booklet provided.
All questions carry equal marks.
Candidates should answer the questions in English.*

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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1. (a) Distinguish between a *master file* and a *transaction file* as used in computer systems. (4 marks)
- (b) Explain **three** advantages of using *database management systems* in an organization. (6 marks)
- (c) Describe each of the following as applied in project management:
 - (i) Optimistic time;
 - (ii) Most likely time;
 - (iii) Pessimistic time. (6 marks)
- (d) PERT is used to review the progress of projects during project management. Explain **two** limitations of this technique. (4 marks)
2. (a) Define the term *early warning mechanism* as used in management information systems. (2 marks)
- (b) The questionnaire is a method used for data collection. Explain **three** disadvantages of using this method during data collection. (6 marks)
- (c) Dorothy intends to design user documentation for the program she developed. Explain **three** qualities of such a documentation that she should consider. (6 marks)
- (d) Most organizations require their new employees to be trained on the use of management information systems as part of employee orientation. Explain **three** reasons for this type of training. (6 marks)
3. (a) Define the term *milestone* as used in project management. (2 marks)
- (b) Explain a circumstance that would necessitate a systems analyst to recommend each of the following types of system maintenance:
 - (i) corrective maintenance;
 - (ii) perfective maintenance;
 - (iii) adaptive maintenance. (6 marks)
- (c) Most organizations have their management information systems connected to a Local Area Network. Explain **three** benefits such organizations are likely to enjoy. (6 marks)
- (d) Capita Company intends to use the *Parallel changeover* method to implement its new information system.
 - (i) Explain **two** advantages of this method to the company;
 - (ii) Explain one disadvantage of using this method. (6 marks)
4. (a) Outline **four** ways in which a *Transaction Processing System* improves operation in an organization. (4 marks)
- (b) Managers rely on *decision support systems* to assist them in decision making. Describe **three** components of this system. (6 marks)

- (c) Implementation of information systems in organizations may sometimes fail. Explain **three** measures that could be taken to prevent such failures. (6 marks)
- (d) Management information systems could be classified according to the type of support provided. Describe this type of classification. (4 marks)
5. (a) Outline **four** features of an *executive information system*. (4 marks)
- (b) Describe each of the following types of systems:
- (i) office automation system;
 - (ii) GIS;
 - (iii) virtual reality. (6 marks)
- (c) Distinguish between *structured* and *unstructured* decisions. (4 marks)
- (d) Debris Company has installed a biometric system to control access to its offices. Explain **three** examples of biometric data that could be used in such a system. (6 marks)
6. (a) Information technology has been considered as an agent of organizational change. Explain **three** ways in which it has been applied to bring about this change. (6 marks)
- (b) The following features describe a certain method used during the procurement process in an organization. Use them to answer the questions that follow:
- There is only one person or entity who can supply the goods or services being procured
 - There is no reasonable alternative or substitute for the goods or service.
 - There is an urgent need for the goods or services being procured
- (i) Identify the method of procurement described by these features. (1 mark)
- (ii) Outline **three** disadvantages of this method. (3 marks)
- (c) Explain **two** reasons that would necessitate an organization to embrace *cloud computing* technology in their operations. (4 marks)
- (d) Describe the *decision making process*, using the Herbert Simons' model. (6 marks)
7. (a) Explain **two** reasons for managing *information resources*. (4 marks)
- (b) Computer users sometimes engage in unethical use of information in the course of duty. Explain **three** such unethical behavior. (6 marks)
- (c) Distinguish between *a hacker* and *a cracker* as used in computer crime. (4 marks)
- (d) With the aid of a diagram, outline the steps involved in *information system planning*. (6 marks)
8. (a) Define the term *guarantee* as applied in management information systems. (2 marks)
- (b) Management information systems assist many organizations in marketing their products and services. Explain **three** ways in which an information system can be used to achieve this. (6 marks)

- (c) Computers are prone to attacks by viruses from various sources.
- (i) outline **three** types of computer viruses; (3 marks)
 - (ii) outline three ways of protecting computer systems against such viruses. (3 marks)
- (d) The management of a certain company has acquired a new management information system for use and some staff are resisting this move. Other than training, explain **three** measures the company could adopt to minimize the resistance. (6 marks)

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